

ATROPOS HOE UI/UX GAP MAP v2

Authority: Source Document 4 + HOE 100% Completeness Handoff v2 + 29 Jul 2026 export + Antigravity-class CLI directive Density: ultra-compressed atoms · fingerprints · predicates · edges · shall · RESEARCH · IMPL Core Engine Gap Map: separate · Phase 20: separate Job Hunter Pro: method inspiration only — zero product content mixed

0. AUTHORITY LEDGER

- Source Doc 4 v1.0 + Handoff v2 (2026-07-29)
 - Export: file_count=1475 · generated_utc=2026-07-29T13:24:57Z
 - CLI paths: src/main/kotlin/atropos/cli/ui/** · LandingRenderer · HomeStateProvider · AnsiTerminalEngine · CommandRouter · DesignTokens
 - Web: apps/atropos-web/ (Phase 0) · packages/atropos-web-contracts/
 - Android: absent HOE chrome · bridge required
 - Screenshot baseline: current CLI (provider wall, mythology answers, no sticky chrome, no progressive disclosure)
 - Competitive targets: CLI=Antigravity-class · Web=OpenCode-class · Android=Claude-density under HIG
 - Success metric: every engine capability that requires a surface has a completed HOE endpoint atom
1. SHARED FOUNDATION (A-series) HOE-A01 | Six continuous answers on every primary surface without search | PARTIAL CLI | PARTIAL | all RESEARCH: Map LandingRenderer/HomeStateProvider fields to the six questions; list missing answers on current CLI screenshot. IMPL: Extend existing status/home renderers only; never invent a parallel status system.

HOE-A02 | Primary navigation spine

Home·Projects·Work·Conversations·Files·Agents·Models·Automation·History·Settings·Developer Tools (hidden) | PARTIAL | PARTIAL | all RESEARCH: Diff current CommandRouter/help registry against Source Doc 4 spine; identify absent entries. IMPL: Register spine items in existing command registry; Developer Tools remain hidden by default.

HOE-A03 | Project is durable organizational boundary; restart-safe | PARTIAL | PARTIAL | all RESEARCH: Locate existing project/session persistence paths; measure what survives JAR restart today. IMPL: Bind UI project identity to existing durable stores; no new project database.

HOE-A04 | Status vocabulary user-progress oriented + non-color | PARTIAL | PARTIAL | all RESEARCH: Extract current status strings from DesignTokens/renderers; check color-only reliance. IMPL: Adopt Source Doc 4 vocabulary (Idle/Planning/Waiting/Working/Review Required/Blocked/Completed/Failed/Cancelled); pair every color with icon+text.

HOE-A05 | Evidence affordance on every completion claim | PARTIAL | PARTIAL | all RESEARCH: Find all completion claim render sites; list which lack evidence link. IMPL: Add evidence affordance to existing completion cards; reuse EvidenceCollector paths.

HOE-A06 | Why/How/Evidence actions on significant recommendations | PARTIAL | PARTIAL | all RESEARCH: Locate recommendation/agent-output surfaces; measure current explainability controls. IMPL: Wire Why/How/Evidence to existing verifier/evidence stores; progressive disclosure only.

HOE-A07 | Command palette reaches every primary action | PARTIAL | PARTIAL | all RESEARCH: Diff CommandPaletteRenderer registry against all primary actions in Source Doc 4. IMPL: Generate palette from single command registry; never hard-code entries.

HOE-A08 | Progressive disclosure L1-L4; information never removed | PARTIAL | PARTIAL | all RESEARCH: Inventory current disclosure levels in CLI/Web; identify any information-removal paths. IMPL: Default collapsed; each expand reveals only additional detail; never hide prior level.

HOE-A09 | Workspace layout persistence + recovery report | PARTIAL | PARTIAL | all RESEARCH: Inspect CrashRecoveryService / StateSnapshot surfaces; what is reported on restart today. IMPL: Surface recovery ribbon from existing restart coordinator; no new recovery engine.

HOE-A10 | Secret-safe rendering; no raw secrets in ordinary views | PARTIAL | PARTIAL | all RESEARCH: Grep render paths for secret leakage; confirm RedactionFilter call sites before UI. IMPL: All renderers call redaction before paint; secrets never appear in ordinary views.

2. CLI/TUI — ANTIGRAVITY-CLASS (B-series) HOE-B01 | Sticky header (brand · ContextSight pill · checkpoint age) + anchored bottom input; no bounce on redraw | screenshot fails | OPEN | B02 RESEARCH: Capture live Antigravity (agy) header/chrome regions and keybind map; measure redraw cost on aarch64 Termux. IMPL: Extend LandingRenderer/HomeStateProvider sticky regions; keep AnsiTerminalEngine as sole canvas owner.

HOE-B02 | Disclosure rows default collapsed: ▶ Thinking · ▶ Plan · ▶ Evidence · ▶ Engine · ▶ Checkpoint | missing | OPEN | B03 RESEARCH: Prototype ▶ row expand/collapse cost at 40/80/120/160 cols; confirm reduced-motion path. IMPL: Add disclosure rows to existing transcript renderer; default collapsed; L1 only on first expand.

HOE-B03 | Multi-level thinking L1 outline → L2 → L3 explicit expand; independent surface verbosity | missing | OPEN | B04 RESEARCH: Define L1/L2/L3 payload schema from existing agent journals; ensure independent channels from Web/Android. IMPL: Thinking depth is UI filter only; never change provider task payload; store full depth in engine, surface on demand.

HOE-B04 | Checkpoint chip + resume panel (StateSnapshot presentation); one primary continue from checkpoint | PARTIAL engine | OPEN | B05 RESEARCH: Map .atropos/handoffs/ and existing StateSnapshot fields to chip + resume panel schema. IMPL:

Present existing checkpoint as product object; one-shot restore goal/node/pack; no re-prompt mythology.

HOE-B05 | Partial-command Enter-to-select (type /pro.. → highlight → Enter selects) | current forces full retype | OPEN | B06 RESEARCH: Inspect current PromptState/completion list; measure Enter-to-select vs retype friction. IMPL: Wire existing suggestion list to Enter accept; never require full retype of highlighted command.

HOE-B06 | Providers default one-line healthy summary; full matrix only on /providers full or expand | screenshot shows wall | OPEN | B07 RESEARCH: Reduce current provider dump to one-line healthy summary; define expand trigger. IMPL: Default /providers = compact healthy line; full inventory only on explicit expand or /providers full.

HOE-B07 | Animated thinking buffer (spinner/progress on running chip only); ~30 FPS; reduced-motion safe | missing | OPEN | B08 RESEARCH: Frame-budget test spinner vs full repaint on aarch64; define reduced-motion fallback. IMPL: Animate only running chip; never stream engine logs into main transcript; respect prefers-reduced-motion.

HOE-B08 | Copy/download response from within CLI (select + clipboard) | missing | OPEN | B09 RESEARCH: Termux clipboard APIs and existing copy paths; define selectable response card bounds. IMPL: Add copy action to response cards; reuse existing clipboard bridge if present.

HOE-B09 | Territory-as-material + attestation optical focus (weight/spectral edge) | missing | OPEN | B10 RESEARCH: Bind territory in/out and attestation validity to existing DesignTokens weight/color roles. IMPL: Out-of-territory desaturates; valid attestation sharpens type; no decorative glow without state.

HOE-B10 | Responsive native grammar COMPACT(<60)·MEDIUM·WIDE(100+); same roles; NO_COLOR scannable; resize-safe input | DesignTokens exist | PARTIAL | — RESEARCH: Golden snapshots at 40/80/120/160 cols; TERM=dumb output; pinch-zoom reflow on Termux. IMPL: Extend existing DesignTokens breakpoints; preserve prompt/cursor across resize; never clip brand.

HOE-B11 | Baseline snapshots 40/80/120/160 cols no overflow; golden TERM=dumb | partial | PARTIAL | — RESEARCH: Refresh docs/ui-parity/baseline/ with current and target renders. IMPL: Store fingerprints; acceptance = no overflow + scannable NO_COLOR.

3. WEB — OPENCODE-CLASS (C-series) HOE-C01 | Scaffold ATROPOS Web shell per WEB_MERGE_ARCHITECTURE; local-first | Phase 0 only | OPEN | C02 RESEARCH: Read WEB_MERGE_ARCHITECTURE.md; list missing shell routes vs OpenCode session model. IMPL: Own only apps/atropos-web/; SpecGraph stays under /developer/specgraph.

HOE-C02 | Home + Projects + Work answer six questions | missing | OPEN | C03 RESEARCH: Map six questions to Web home cards; reuse CLI status contracts. IMPL: Thin presentation over existing engine status endpoints; no business logic in Web.

HOE-C03 | Session/tab model (OpenCode parity); workspace survives refresh | missing | OPEN | C04 RESEARCH: OpenCode session restore UX vs ATROPOS StateSnapshot; define tab persistence schema. IMPL: Local-first session/tab; survive refresh via existing durable stores.

HOE-C04 | Checkpoint rail/header chip + side panel (summary/next/evidence) | missing | OPEN | C05 RESEARCH: Design checkpoint as product object on Web; resume primary, not new-chat. IMPL: Same StateSnapshot presentation as CLI; independent verbosity channel.

HOE-C05 | Streaming + approval cards + command palette | missing | OPEN | C06 RESEARCH: SSE/event stream contracts already exposed; approval card schema for APPROVAL_REQUIRED. IMPL: Consume existing bridge/events; never reimplement policy.

HOE-C06 | Multi-level thinking / Evidence / Engine expandable in side panel or drawer | missing | OPEN | C07 RESEARCH: Side-panel vs drawer density for L1-L3; independent of CLI channel. IMPL: Collapsed default; expand on demand; full depth stored in engine.

HOE-C07 | Developer Tools container; SpecGraph only under /developer/specgraph | missing | OPEN | — RESEARCH: Confirm SpecGraph mount path remains isolated. IMPL: No SpecGraph in primary nav; Developer Tools hidden by default.

HOE-C08 | Copy/download response; evidence morph (View Transition card → drawer) | missing | OPEN | — RESEARCH: View Transitions API support; evidence drawer field list from Source Doc 4. IMPL: Morph only when evidence exists; bind motion to real state.

HOE-C09 | Shared contracts in packages/atropos-web-contracts | partial | PARTIAL | — RESEARCH: Inventory existing contracts vs required bridge API. IMPL: Extend contracts package; all surfaces consume same types.

4. ANDROID APK — CLAUDE-DENSITY (D-series) HOE-D01 | App shell (chat list · conversation · composer · engine offline) | absent | OPEN | D02 RESEARCH: Claude Code mobile density patterns; one-hand reach zones under Android HIG. IMPL: Thin native (Compose) or WebView client; never embed full engine.

HOE-D02 | Local engine bridge (session/message/events/status/approve/evidence/files/cli) | absent | OPEN | D03 RESEARCH: Minimal bridge API latency on device; binder vs localhost trade-off. IMPL: Implement once; all Android surfaces consume it; full power only via bridge.

HOE-D03 | Primary column stream + secondary tools/timeline | absent | OPEN | D04 RESEARCH: Claude-web column split adapted to phone width; secondary as sheet. IMPL: Primary stream always visible; secondary sheet for tools/timeline.

HOE-D04 | Checkpoint chip under answer or status bar; tap → sheet (Resume/Next/Evidence) | absent | OPEN | D05 RESEARCH: Checkpoint chip placement for one-hand use; sheet content schema. IMPL: Same StateSnapshot as CLI/Web; resume is primary action.

HOE-D05 | Multi-level thinking expandable inside sheet; independent of terminal | absent | OPEN | D06 RESEARCH: Sheet expand cost; independent verbosity from CLI. IMPL: Collapsed default; L1–L3 on demand; never force deep thinking on mobile.

HOE-D06 | Approval cards for APPROVAL_REQUIRED; push on approval-needed | absent | OPEN | D07 RESEARCH: Android notification patterns for approval; push permission timing. IMPL: Approval cards in stream; push only for APPROVAL_REQUIRED; never silent.

HOE-D07 | One-hand density, HIG targets, offline project resume | absent | OPEN | D08 RESEARCH: 44×44 pt targets; offline resume from local store. IMPL: Meet HIG; resume from durable project state without laptop.

HOE-D08 | Copy/share response; full CLI power via bridge | absent | OPEN | D09 RESEARCH: Share sheet vs clipboard; bridge passthrough for CLI commands. IMPL: Copy/share from response; full CLI only through bridge, not reimplementation.

HOE-D09 | Sideload-ready artifact | absent | OPEN | — RESEARCH: APK signing + sideload path on target devices. IMPL: Produce sideloadable APK; document install steps.

5. CONTINUOUS SUPERIORITY + CROSS-SURFACE (E/F) HOE-E01 | Territory-as-material (in/out visual weight) | OPEN RESEARCH: Bind territory state to DesignTokens desaturation; test across surfaces. IMPL: Out-of-territory recedes; in-territory holds accent; no parallel visual system.

HOE-E02 | Attestation-as-optical-focus (type weight / spectral edge) | OPEN RESEARCH: Variable-font weight or ANSI intensity for attestation validity. IMPL: Valid envelope sharpens; drift softens; certainty readable without numbers.

HOE-E03 | Animated thinking buffer bound to real node progress | OPEN RESEARCH: Node progress fields already in journals; map to chip animation only. IMPL: Animate only when real progress exists; never fake activity.

HOE-E04 | Independent surface verbosity channels (terminal deep / web quiet or reverse) | OPEN RESEARCH: Define channel isolation contract so one surface expand never forces another. IMPL: Each surface owns its disclosure state; engine stores full depth once.

HOE-E05 | Evidence morph / drawer on all surfaces | OPEN RESEARCH: View Transition on Web; sheet on Android; expand row on CLI. IMPL: Same evidence fields; surface-specific gesture only.

HOE-E06 | Recovery tectonic ribbon reports what restored | OPEN RESEARCH: Existing restart report fields; surface on first paint after recovery. IMPL: Ribbon from RestartCoordinator; never silent resume.

HOE-E07 | Mode retheme from real status vocabulary only | OPEN RESEARCH: Map status vocabulary to theme tokens; no random retheme. IMPL: Planning/Working/Review/Blocked drive tokens; never decorative mode switch.

HOE-F01 | Same project identity + status vocabulary on CLI/Web/Android | OPEN RESEARCH: Contract test that status strings and project IDs match across surfaces. IMPL: Single source of truth in engine; surfaces are pure presentation.

HOE-F02 | Delta register updated (changed rows only; fingerprints refreshed) | OPEN RESEARCH: Current HOE_DELTA_REGISTER.tsv location and schema. IMPL: After every batch update only changed rows; store path fingerprints.

HOE-F03 | Competitive checklist green (Antigravity CLI · OpenCode web · Claude Android) | OPEN RESEARCH: Final side-by-side checklist against live competitors. IMPL: Checklist is acceptance gate, not a new feature set.

HOE-F04 | Accessibility: keyboard-complete CLI · non-color status · reduced-motion · high-contrast · screen-reader · HIG targets | OPEN RESEARCH: NO_COLOR golden; screen-reader labels for disclosure rows; contrast on glass. IMPL: Accessibility is release-blocking; pair every color with non-color channel.

6. ORDERED EXECUTION DAG HOE-A01...A10 → HOE-B01...B11 (CLI) → HOE-C01...C09 (Web) → HOE-D01...D09 (Android) → HOE-E01...E07 → HOE-F01...F04 → 100% HOE GATE GREEN Desktop deferred until CLI+Web+Android green.
7. RAW CLOSURE METRICS Shared foundation: PARTIAL · CLI Antigravity: OPEN · Web OpenCode: OPEN · Android Claude-density: OPEN · Continuous: OPEN Every engine capability requiring a surface has a corresponding HOE endpoint atom above.
8. HANDOFF BLOCK AUTHORITY: User directive 2026-07-29 — CLI=Antigravity-class; Web=OpenCode; Android=Claude-density; multi-level thinking + checkpoint optional on every surface. NON-GOALS: Do not redesign web as Antigravity; do not make engine logs the home view; do not leave thinking expanded by default; do not mix Job Hunter Pro content. Every atom above carries RESEARCH (one-line LLM prompt) and IMPL (1–2 line necessary note). Pursue RESEARCH before coding when gap is OPEN. IMPLEMENTATION ORDER: A → B → C → D → E/F DELIVERABLE LANGUAGE: Prefer shall/must/evidence/acceptance; attach path fingerprints; update only changed delta-register rows.

End of HOE gap map v2.